Operations Research (OR) TD 1

OR and modeling art 3RD YEAR - ENSIA

Exercise C1

A man holds 3000 Apples in storage and he wants to sell them in the market which is 1000 KM away. The only way he can transport these apples is via a Donkey. This Donkey holds an insatiable appetite because it eats one apple every kilometer it walks. In addition, the donkey can **carry a maximum of 1000 apples at a time**. The goal is to determine the **maximum number of apples that can be delivered to the market**.

(a) What is the **criterion of optimization** here?
(b) What is the maximum number of apples that can arrive at the market?
(c) What makes the problem difficult?

Exercise C2

A small company must perform **4 tasks** to produce a product. Tasks have durations and precedence constraints:

- Task **A:** duration **4** time-units.
- Task **B:** duration **3** time-units. (can start **only after A** is finished)
- Task **C:** duration **2** time-units. (can start **immediately**, no prerequisite)
- Task **D:** duration **5** time-units. (can start **only after both B and C** are finished)

(a) Describe the precedence rules in plain sentences.
(b) Draw a feasible Gantt chart for the case with **one machine** (only one task at a time).
(c) Compute the makespan (total project duration) for the one-machine schedule.
(d) Identify the sequence of dependent tasks that forces the earliest possible project completion time.
(e) Now assume **two identical machines** (two independent tasks can run at the same time). Propose an optimized schedule, draw a Gantt chart, compute the makespan, and explain why it is optimal.

Exercise C3

There are 100 matchsticks. Two players take turns. On each turn, a player can take 1, 2, or 3 sticks. The player who takes the last stick wins.

(a) What is the **decision criterion** ?

(b) What is the winning strategy?

Exercice C4

You have a 3-liter jug and a 5-liter jug. You need exactly 4 liters of water. The jugs have no markings other than full/empty. Find a sequence of steps to measure 4 liters.

Exercise P1 (Optional)
You have five chains, each consisting of four solid links. You need to make a bracelet by connecting all five chains. It costs 2 cents to break a link and 3 cents to re-solder it.
(a) Identify two feasible solutions and evaluate them.
(b) Determine the cheapest cost for making

Exercise P2 (Optional)
In a baseball game, Jim is the pitcher and Joe is the batter. Suppose that Jim can throw either a fast or a curve ball at random. If Joe correctly predicts a curve ball, he can maintain a .400 batting average, else, if Jim throws a curve ball and Joe prepares for a fast ball, his batting average is kept down to .200. On the other hand, if Joe correctly predicts a fast ball, he gets a .250 batting average, else, his batting average is only .125.

(a) Define the alternatives for this situation.
(b) Define the objective function for the problem and discuss how it differs from the familiar optimization (maximization or minimization) of a criterion.

Exercise P3 (Optional)
You have 10 identical cartons each holding 10 water bottles. All bottles weigh 10 oz. each, except for one defective carton in which each of the 10 bottles weighs on 9 oz. only. A scale is available for weighing.

(a) Suggest a method for locating the defective carton.
(b) What is the smallest number of times the scale is used that guarantees finding the defective carton? (Hint: You will need to be creative in deciding what to weigh.)

Exercise P4 (Optional)
You are given two identical balls made of a tough alloy. The hardness test fails if a ball dropped from a floor of a 120-storey building is dented upon impact. A ball can be reused in fresh drops only if it has not been dented in a previous drop. Using only these two identical balls, what is the smallest number of ball drops that will determine the highest floor from which the ball can be dropped without being damaged?